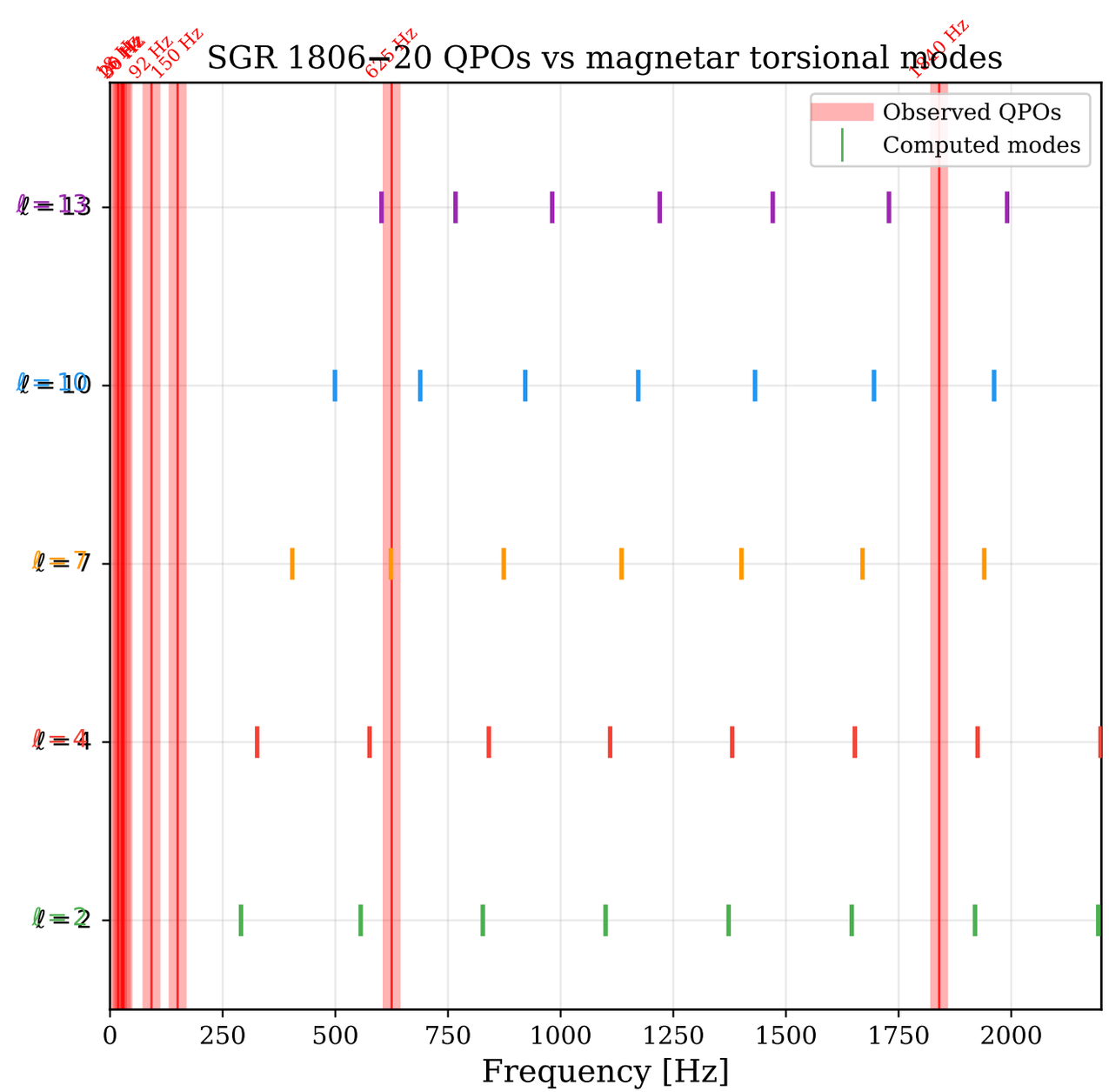
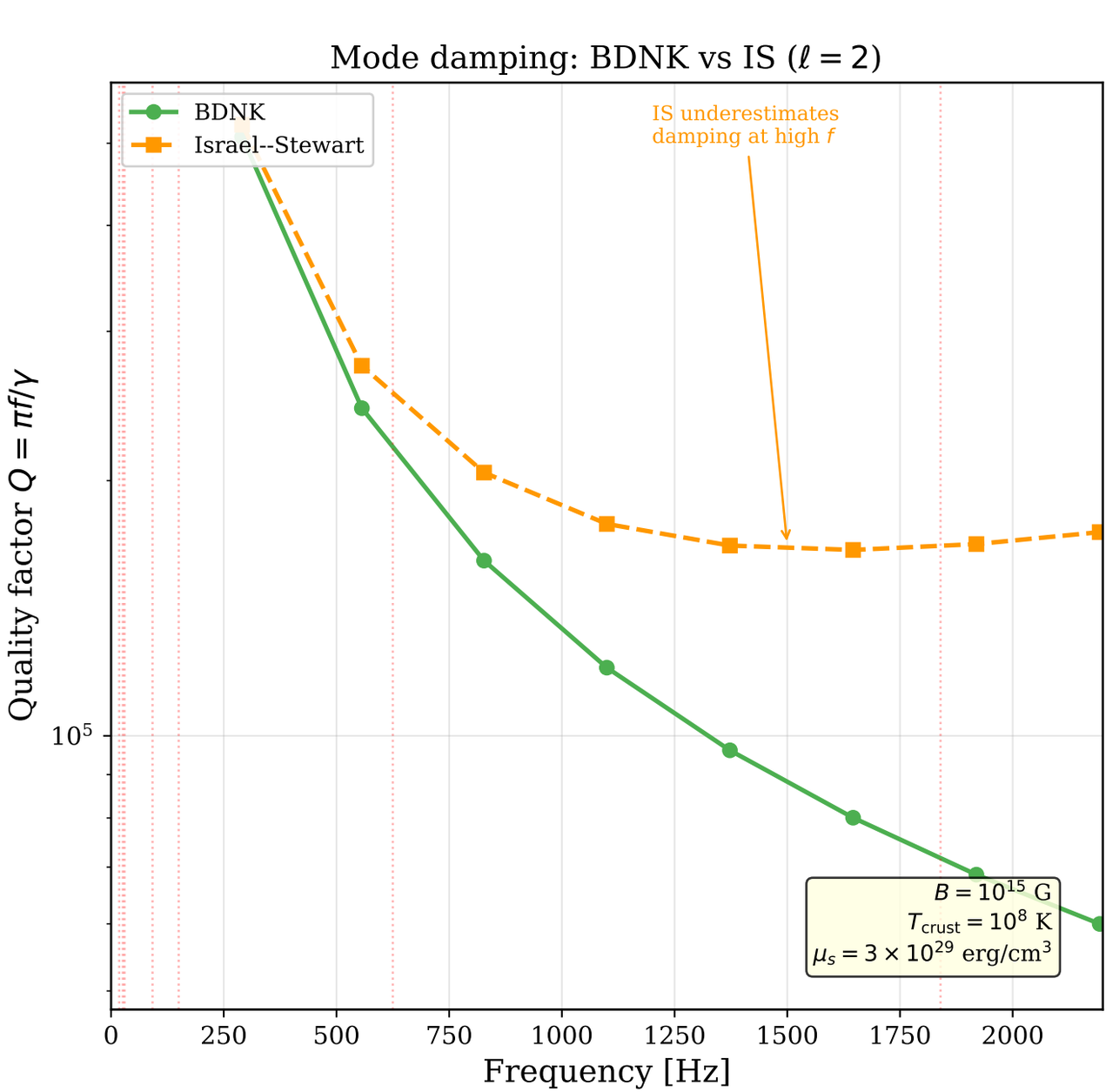

SGR 1806−20 QPOs vs magnetar torsional modes
Observed QPOs
Computed modes
ℓ = 13
ℓ = 10
ℓ = 7
ℓ = 4
ℓ = 2
0
250
500
750
1000
1250
1500
1750
2000
Frequency [Hz]


Mode damping: BDNK vs IS (ℓ = 2)
BDNK
Israel--Stewart
IS underestimates
damping at high f
Quality factor Q = πf/γ
10^5
B = 10^15 G
T_crust = 10^8 K
μ_s = 3 × 10^29 erg/cm^3
0
250
500
750
1000
1250
1500
1750
2000
Frequency [Hz]